

Statement of Work & Requirements

Introduction:

This procurement is for processing (melting and casting) of experimental NiTi-16Hf alloys. Final form should be rod/bar stock (1" - 1.125" diameter) and lengths of approximately 6". As these are experimental alloys, there are no commercial standards or specifications to which the material can be stated. The rods **MUST** be in one piece with no cracks.

Statement of Work:

For all items (3) we would like to procure vacuum arc melted, cast rod (1" - 1.125" diameter) of the composition and quantity defined below:

Item 1.) **0.6 to 0.8 kg** of the following alloy in atomic percent (at.%): **50.6Ni – 33.4Ti – 16Hf**; cast into ~1" rod. No other secondary processing needed.

Requirements:

1. The supplier shall have demonstrated experience in the melting of NiTiHf shape memory alloys with good quality. Alloys should be clean, homogeneous, and with very low inclusion content, Impurities, such as O, and C will need to be at kept at minimal levels. **Companies that have not supplied material previously for our application should provide a 3 kg. test/evaluation charge of item #1 upon request for evaluation at GRC.**
2. A single heat will be prepared for each item (alloy) above, formulated using virgin melt stock. A certificate of chemical analysis will be supplied for each master heat.
3. Melting to be performed under vacuum to minimize oxide inclusion content. Every effort should be made to minimize ceramic or other inclusions and ensure that the melt is homogeneous without unmelted metal content.